

CommCanary 0.4v: Fail Closed Evidence for Decision Fidelity

Evaluation of Collective Trace Replay.

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Abstract

Collective communication traces can replace expensive distributed workloads in controlled experiments, but replaying the right calls does not guarantee the same configuration choice. CommCanary provides a fail closed workflow for freezing, selecting, archiving, and independently checking collective trace replay evidence. Its publication path binds a campaign to immutable attempts, an explicit selection, a completeness verdict, a hash identified archive, and deterministic analysis. Missing or unstable evidence withholds the claim.

We compare four replay representations with a full workload on one node with four NVIDIA A100 GPUs and eight NCCL configurations. The trusted join contains all 280 selected cells. Communication only replay does not preserve the measured workload decision. W-canary reaches 57.1% pair agreement, below the 64.3% of W-micro. Adding compute and communication concurrency changes the ranking. W-shared-overlap reaches 71.4% agreement and $\tau_b = 0.708$, but still disagrees on 8 of 28 pairs.

A separate exact work gate reports 26 of 28 agreeing pairs, or 92.86%, with $\tau_b = 0.857$, one false negative, one false positive, 1.55% median absolute relative error, and 4.05% p95 error. The persisted verdict is *inconclusive*, not pass. Bootstrap intervals cross frozen decision boundaries, and one configuration exceeds the relative IQR stability limit. The evidence supports a measured negative result about communication only replay and an independently verifiable publication workflow. It does not qualify CommCanary as a production gate, prove general decision preservation, or establish savings from a reduced physical canary.

1 Introduction

Distributed machine learning and high performance computing systems often require a choice among collective algorithms, protocols, or library versions. That choice depends on more than message size. Process arrival times, network topology, endpoint resource use, operation order, and overlap with computation can all change the ranking. MPI implementations therefore select among multiple collective algorithms, and NCCL exposes algorithm and protocol choices whose performance depends on workload context [1–4]. A microbenchmark can measure a collective precisely while omitting the execution conditions that determine the application choice.

Execution traces provide another way to study a workload. A trace can record operations and dependencies; a simulator or replay engine can then execute the recorded structure without rerunning the original application [5–10]. The relevant fidelity

target still has to be stated. A tuning workflow usually asks which configuration should be selected. Small timing error is not enough if the replay reverses two close configurations. High point agreement is also insufficient when the measurements are unstable or uncertainty spans the decision boundary.

This work asks whether collective trace replay preserves the pairwise configuration decisions of a full workload, and whether the evidence path refuses a positive claim when completeness, uncertainty, or stability is inadequate.

CommCanary handles the evidence question first. A campaign is frozen before execution. Each expected cell writes immutable attempts. A separate selection identifies one terminal attempt for each cell. A completeness verdict checks that the selected matrix is present, unambiguous, and successful. The raw evidence is archived and hashed. Publication files are generated deterministically from that selection. A failed boundary stops the claim path.

We use this workflow in two experiments. The first joins three complete campaigns into 280 selected cells that cover eight NCCL configurations and seven workload profiles. It compares four replay representations with the full workload. Communication only W-canary agrees on 16 of 28 configuration pairs, or 57.1%. The isolated W-micro agrees on 18, or 64.3%. A shared trace replay with concurrent compute reaches 20, or 71.4%. Concurrency changes the measured ranking, but no proxy preserves all decisions.

The second experiment reconstructs the source derived work exactly and evaluates it with a policy frozen before analysis. Exact work reaches 26 of 28 pair agreement with low relative timing error. Every point criterion passes. The gate still returns *inconclusive* because pair intervals cross tie boundaries and the `nccl-2.20.5-tree-11` configuration is unstable in three representations. A favorable estimate is therefore recorded without being promoted to a supported qualification claim.

The paper contributes four items:

1. a fail closed evidence model for physical collective replay experiments, with immutable attempts, explicit selection, completeness, archive binding, and deterministic publication;
2. a complete 280 cell comparison showing that faithful communication only replay is insufficient for the measured decision and that compute and communication concurrency changes the ranking;
3. a predeclared exact work gate that reports strong point estimates and the final inconclusive verdict; and
4. an explicit separation between publication evidence, diagnostic evidence, incomplete integration records, and work that has not run.

All physical observations come from a single node with four A100 GPUs. The results do not establish multi-node behavior, expert parallel replay, congestion fidelity, CUDA graph fidelity, vendor qualification, production CI readiness, or runtime savings from a reduced canary.

2 Background and problem definition

2.1 Collective configuration depends on context

Collective operations define group communication semantics while leaving implementation strategy open [1]. Classic MPI work shows that no single algorithm dominates across message sizes, process counts, and systems [3, 4]. NCCL provides topology aware GPU collectives and allows algorithm and protocol selection [2]. Our configuration set contains two NCCL versions at default settings and six choices formed by crossing Ring or Tree with LL, LL128, or Simple on NCCL 2.20.5.

An isolated microbenchmark is useful for transport characterization and regression isolation. It intentionally removes application context. Concurrent compute and communication can contend for endpoint resources and change the critical path [11]. Arrival skew, explicit waits, dependencies, and buffer lifetimes can also change which configuration is preferred. The microbenchmark ranking is therefore an experimental result, not a default model of the full workload ranking.

2.2 Trace replay and surrogate workloads

Trace systems support different goals. ScalaTrace compresses and replays communication traces [9]. LogGOPSIm combines traces with an analytical communication model [10]. ASTRA-SIM explores distributed training systems through workload and network models [7, 8]. Chakra defines an execution trace schema for operations and dependencies, while Mystique derives portable AI benchmarks from execution traces [5, 6]. PARAM supplies communication benchmark and replay tooling used by the experiment path [12].

Trace portability does not decide the claim target. CommCanary tests decision fidelity: whether a replay supports the same configuration choice as the source workload under a frozen policy. The contribution is not a new trace format or simulator. It is a decision level evaluation coupled to a physical evidence boundary that can withhold the claim.

2.3 Decision fidelity

Let C be the measured configurations and R a representation such as the full workload, a microbenchmark, or a replay. For each unordered pair (a, b) , a policy maps timing distributions to

$$d_R(a, b) \in \{a < b, a \sim b, b < a\}, \quad (1)$$

where $<$ means faster and $\sim$ means an operational tie. Agreement between representation R and the full workload F is

$$A(R, F) = \binom{|C|}{2}^{-1} \sum_{a < b} \mathbf{1}[d_R(a, b) = d_F(a, b)]. \quad (2)$$

Eight configurations produce 28 pair decisions. We also report Kendall’s tie adjusted statistic [13]:

$$\tau_b = \frac{N_c - N_d}{\sqrt{(N_c + N_d + T_R)(N_c + N_d + T_F)}}. \quad (3)$$

Here N_c and N_d count concordant and discordant pairs, while T_R and T_F count pairs tied on only one side. Exact agreement requires the same three way label. Kendall τ_b discounts some tie disagreements. Reporting both exposes cases where a representation preserves the non-tied order but changes many decisions into ties.

The trusted join uses its analyzer’s frozen IQR rule. A pair is tied when its median difference is smaller than either configuration’s IQR. The exact work gate is a separate experiment with a pair specific tie band, bootstrap intervals, and a stability requirement. Its thresholds are not applied retroactively to the trusted join.

2.4 Point estimates and verdicts

Performance samples can be autocorrelated, multimodal, or sensitive to machine state. Scientific benchmarking practice calls for a complete method, repeated observations, and explicit uncertainty [14]. The exact work gate uses percentile bootstrap intervals [15] and a relative IQR stability limit. A point estimate can pass while its interval overlaps a tie or reversal region. In that case, the policy returns insufficient evidence rather than pass.

3 CommCanary evidence model

3.1 Fail closed publication path

Figure 1 shows the publication path. The boundaries remain separate because each answers a different question.

1. The campaign manifest freezes the matrix, environment identities, producer identities, policies, and expected cells before measurement.
2. Each cell owns an immutable sequence of attempts. A retry appends a new attempt.
3. A selection names the terminal attempt used for every expected cell and is itself hashed.
4. A completeness verdict checks cardinality, identity, terminal status, and declared artifacts.
5. An archive descriptor binds the retained raw evidence to exact bytes.
6. The analyzer emits aggregate JSON and CSV, a paper fragment, and policy verdicts from the selected evidence.

The absence of a completeness verdict is not an empty valid sample. A parser’s ability to compute a number does not authorize publication. Synthetic fixtures test software behavior but do not enter physical claims. Operational notes may explain an attempt, but they cannot replace an aggregate or persisted verdict.



Any missing identity, hash mismatch, incomplete selection, or absent archive stops the claim path.

Fail closed: publication requires every boundary to pass.

Figure 1: CommCanary publication path. Campaign identity, immutable attempts, selection, completeness, archive binding, and publication are separate gates. Missing or inconsistent evidence stops the claim path.

3.2 Evidence grades

Table 1 separates the material used in this paper. Publication evidence has a complete hash bound campaign and generated analysis. Diagnostic evidence supports a narrow observation without issuing a qualification verdict. Failed integrations remain engineering records and are excluded from quantitative results. Product studies that have not run are future work.

3.3 Independent regeneration

The evidence commit retains primary and reproduction directories for the exact work gate. Their `aggregate.csv`, `aggregate.json`, `decision-fidelity-verdict.json`, and `paper-fragment.md` files have identical Git blob identities. This confirms that the checked in deterministic publication path produced the same bytes for the retained inputs. It does not show that physical measurements reproduce on another machine, and it does not turn a focused analyzer check into release readiness.

4 Experimental methodology

4.1 Campaign inventory and hardware scope

Table 2 lists the three evidence packages. The main comparison joins three campaigns from one four GPU A100 node. Eight configurations, seven workload profiles, and five selected repetitions produce 280 cells. Each cell contributes 100 retained samples after warmup. The exact work gate has one cell per configuration and 20 measured iterations for every representation after five warmups. The diagnostic contains one configuration and compares source and replay on the same node.

All claims are conditioned on the measured node, workload, configuration set, and campaign identity. No result is extrapolated to NVSwitch, another accelerator generation, a multi-node fabric, or concurrent cluster traffic.

4.2 Configurations

The finite decision set contains NCCL 2.19.3 with default algorithm and protocol selection, NCCL 2.20.5 with default se-

lection, and NCCL 2.20.5 with Ring or Tree crossed with LL, LL128, or Simple. The campaign identity freezes the versions and environment. These configurations are not a representative sample of all NCCL behavior; they are the eight choices evaluated here.

4.3 Representations

The trusted join contains seven workload profiles. Four are compared directly with the full workload.

W-micro. An isolated collective microbenchmark. It removes workload computation and timing context.

W-canary. A faithful communication only replay of the recorded collective requests and explicit waits. It preserves communication structure but omits concurrent source computation.

W-canary-overlap. A configuration specific overlap replay that places compute around communication according to the captured schedule.

W-shared-overlap. An overlap replay driven by one shared trace. The replayed work remains fixed while the NCCL configuration changes.

The full workload, **W-full**, is the reference. The difference between the overlap representations matters because a configuration specific trace can encode configuration dependent timing. A shared trace controls that source of variation. Neither replay is presented as a reduced physical canary, and the measured campaigns establish no runtime or artifact size advantage over the full workload.

4.4 Trusted join analysis

Before aggregation, the analyzer validates result schemas, selected attempts, campaign identities, and completeness. It computes retained timing distributions for each workload and configuration cell. Configuration order follows generated medians. Pair labels use the frozen IQR tie rule from Section 2.3. We

Table 1: Claim boundary by evidence grade.

Grade	Included material	Permitted interpretation
Publication evidence	280 cell trusted join and 8 cell exact work gate	Supports the measured comparison and gate observations only.
Diagnostic	Same node source and replay timing with deterministic checks	Supports reconstruction debugging; no qualification verdict or general claim.
Failed or incomplete integration	Attempts without a complete hash bound publication package	Excluded from quantitative results. An operational failure is not scientific evidence.
Future work	Real vLLM or SGLang corpus, held out physical canary, additional nodes and congestion studies	Not evaluated and not implied by the present artifacts.

Table 2: Campaign inventory. Complete means that the manifest bound completeness verdict passed.

Package	Hardware scope	Matrix and sampling	Selection	Claim status
Trusted comparison join	One node; 4 A100 GPUs	8 configurations, 7 profiles, 5 selected repetitions per cell, 100 retained samples per cell	280 of 280 cells	Complete comparative evidence
Exact work gate	One node; 4 A100 GPUs	8 configurations, 28 unordered pairs, 20 measured iterations after 5 warmups	8 of 8 cells	Point criteria pass; final verdict inconclusive
Exact work diagnostic	Same node; four ranks	1 configuration, 20 source and replay samples, 32 deterministic checks	1 of 1 cell	Complete diagnostic; no qualification verdict

report the 28 exact pair labels and τ_b . Quantitative tables from the historical paper draft are excluded because their original complete evidence was not retained.

4.5 Exact work decision gate

The exact work experiment evaluates source derived execution, verified exact work materialization, and control representations. Its policy was frozen before evaluation. The domain is a single-node, four A100 program with one inflight all reduce at a time, explicit waits, and source observed contiguous GEMM work.

Every representation has five warmups and 20 measured iterations. The policy uses an independent 95% percentile bootstrap with 2,000 resamples and seed 7411. A configuration is unstable when relative IQR exceeds 20%. For source medians m_a and m_b , the pair tie band is

$$\delta(a, b) = \max(10 \mu s, 0.05 \min(m_a, m_b)). \quad (4)$$

Point criteria require at least 80% pair agreement, τ_b of at least 0.7, no more than one false negative and one false positive, median and p95 absolute relative errors below 10% and 15%, and at least two additional agreeing pairs over each control.

Passing these point criteria is necessary but not sufficient. A pass can be withheld when bootstrap intervals cross decision regions or when the stability condition fails.

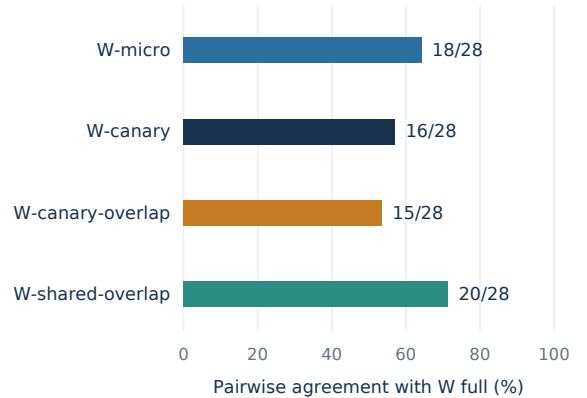
5 Results

5.1 Communication only replay does not preserve the decision

Table 3 and Figures 2 and 3 report the trusted join. W-canary agrees with W-full on 16 of 28 pairs, or 57.1%, with $\tau_b = 0.204$. W-micro agrees on 18 pairs, or 64.3%, with $\tau_b = 0.490$. Replaying the request and wait structure without computation

Table 3: Agreement with W-full over eight configurations.

Representation	Pair agreement	Kendall τ_b
W-micro	18/28 (64.3%)	0.490
W-canary	16/28 (57.1%)	0.204
W-canary-overlap	15/28 (53.6%)	0.677
W-shared-overlap	20/28 (71.4%)	0.708

**Figure 2:** Exact three way pair agreement with the full workload decision. No measured proxy reaches 80%.

is therefore insufficient for this measured decision. Under exact labels it performs worse than the isolated microbenchmark.

The disagreement concerns configuration selection, not only absolute latency. W-canary places the full workload’s worst measured configuration near the front of its ranking. A replay can match call count and byte count while changing the interference and synchronization that produce the ordering.

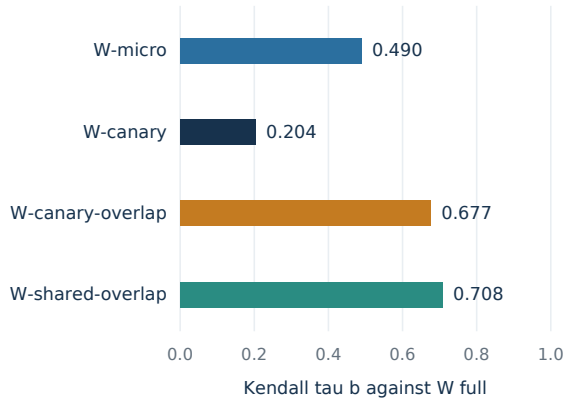


Figure 3: Tie adjusted Kendall correlation with the full workload ordering. Correlation and exact three way agreement answer different questions.

5.2 Concurrency changes the ranking

Adding overlap changes both pair agreement and tie structure. `W-canary-overlap` agrees on 15 of 28 pairs, or 53.6%, while its τ_b is 0.677. The low exact agreement and higher rank correlation are consistent with many pairs becoming policy ties. The remaining non-tied order is more concordant even though the exact labels differ.

`W-shared-overlap` is the strongest measured proxy at 20 of 28 agreement, or 71.4%, with $\tau_b = 0.708$. It gains two agreeing pairs over `W-micro` and four over `W-canary`. Holding the concurrent work constant while changing the NCCL configuration materially changes the result. Eight pair decisions still disagree with `W-full`, so the experiment does not qualify overlap replay.

The trusted join is descriptive and predates the separate exact work policy. We do not apply a new threshold after observing the result. The complete ordering instead motivates the predeclared gate.

5.3 Exact work has strong point estimates

Exact work agrees on 26 of 28 pairs, or 92.86%, with $\tau_b = 0.857$. It records one false negative and one false positive. Median absolute relative error is 1.55%, and p95 error is 4.05%. It gains seven agreeing pairs over the isolated control and sixteen over the stratified control. Every point criterion passes, as Table 4 records.

These estimates show that exact work reconstruction is substantially closer to the source ordering than the controls in this campaign. They do not determine the final claim because a pair decision also requires a supported interval and stable input measurements.

5.4 The predeclared verdict is inconclusive

The persisted outcome is *inconclusive*. Two conditions prevent pass.

First, bootstrap intervals for several exact work pair differences cross the frozen decision regions. Figure 4 normalizes representative differences and 95% intervals by each pair’s tie threshold. An interval below -1 supports the second configu-

ration, an interval above $+1$ supports the first, and an interval contained within $[-1, +1]$ supports a tie. Crossing those regions leaves the label unsupported.

Second, `nccl-2.20.5-tree-11` exceeds the 20% relative IQR limit in source at 47.5%, exact work at 32.3%, and the stratified control at 50.4%. The instability appears in more than the candidate representation. The campaign cannot support a clean qualification involving that configuration under the frozen policy.

The supported reading is asymmetric. Exact work is promising and closer to the source decisions than the controls in this campaign. The evidence does not show that the gate passed, that decision preservation is proven, or that exact work is a qualified production oracle.

5.5 Same node diagnostic

A separate one cell diagnostic compares retained source and replay timing for one exact work capsule. The source median is 1,434.112 μs and the replay median is 1,541.0015 μs . The signed difference is +106.8895 μs , or +7.4533578967%. All 32 deterministic checks pass across four ranks. Table 5 states the complete narrow claim.

The diagnostic confirms deterministic reconstruction checks and a same node timing comparison. It has no multi-configuration full workload matrix. It cannot issue a qualification verdict or substitute for the inconclusive decision gate.

6 Interpretation

6.1 A trace encodes a causal hypothesis

A communication trace selects the workload properties assumed to determine the configuration choice. The communication only representation assumes that collective type, size, order, and synchronization are sufficient. The trusted join rejects that assumption for the measured case. The operation stream can be faithful at the communication interface while omitting concurrent work that changes endpoint contention and critical path timing.

The overlap results narrow the diagnosis. Adding concurrency does not improve every metric. The configuration specific overlap replay has fewer exact matches than the microbenchmark because its tie structure changes. The shared trace variant performs best when concurrent work is held constant across configurations. The overlap schedule, its dependence on configuration, and the tie policy all affect the claim. Including overlap is necessary for this case but is not a complete fidelity recipe.

6.2 Decision fidelity differs from timing fidelity

Absolute error and decision fidelity can diverge. Two configurations may differ by less than measurement variability. A replay can have small percentage error and reverse them, or a common timing bias and preserve their order. We therefore report pair labels, rank correlation, false negative and false positive counts, and timing errors separately.

Table 4: Predeclared exact work point criteria. Every point row passes, but the final verdict also requires supported uncertainty and stability.

Criterion	Observed	Required	Point status
Pair agreement	92.86%	at least 80%	pass
Kendall τ_b	0.857	at least 0.7	pass
False negatives	1	at most 1	pass
False positives	1	at most 1	pass
Median absolute relative error	1.55%	at most 10%	pass
p95 absolute relative error	4.05%	at most 15%	pass
Agreement advantage over isolated	7 pairs	at least 2 pairs	pass
Agreement advantage over stratified	16 pairs	at least 2 pairs	pass

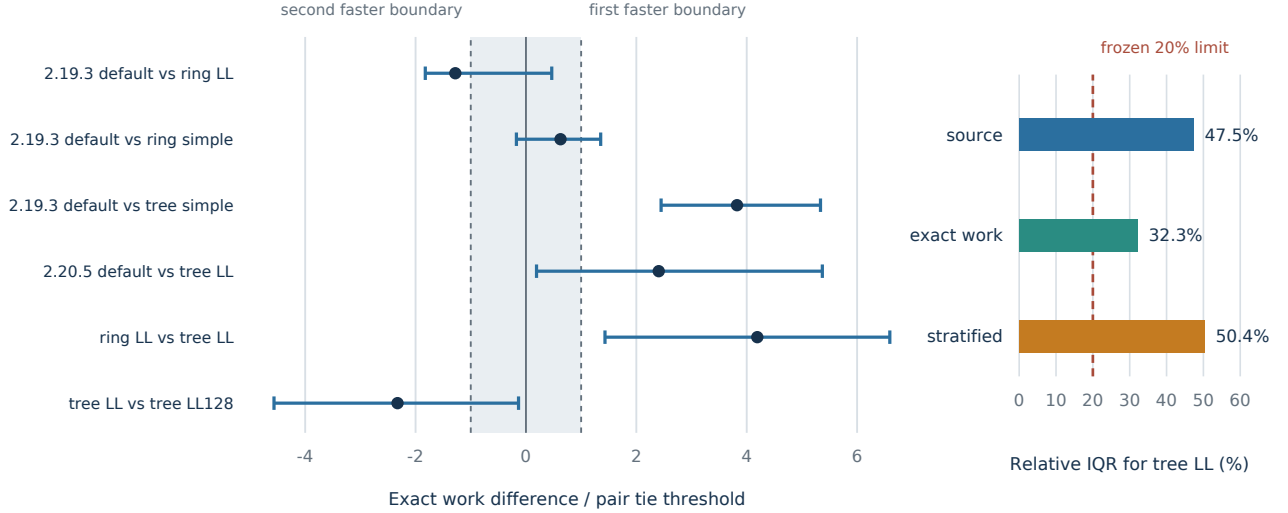


Figure 4: Why the exact work verdict is inconclusive. Representative 95% intervals cross decision regions, and the tree LL configuration violates the frozen stability limit.

Table 5: Exact work diagnostic. No qualification verdict is issued.

Item	Result
Source median	1,434.1120 μ s
Replay median	1,541.0015 μ s
Signed difference	+106.8895 μ s (+7.4534%)
Source / replay IQR	9.2160 / 8.9725 μ s
Deterministic checks	32 passed across 4 ranks
Claim status	diagnostic; physical fidelity unproven; verdict not issued

This distinction explains the exact work outcome. The low median and p95 errors are favorable, but the gate asks whether each operational decision is supported under uncertainty. A 92.86% point agreement is not a 92.86% probability that the replay is correct. The interval can invalidate the point label.

6.3 Fail closed evidence changes the default

Experiment scripts often make it easy to produce a number. Missing runs can disappear during aggregation, and retries can overwrite failures. A fail closed path makes incompleteness visible, preserves failed attempts, and requires explicit selection before analysis. It adds operational work and metadata, but it lets a later reader distinguish a complete negative result from

an incomplete favorable subset.

The exact work verdict is the direct example. A threshold only summary would report that every numeric row passed. Because uncertainty and stability are policy conditions, the retained conclusion is *inconclusive*.

6.4 Unsupported conclusions

The evidence does not show that a reduced physical canary is cheaper than the full workload. The replay proxies are not a completed product corpus, and no held out physical regression study is present. The checked in publication package has no real vLLM or SGLang application campaign. Exact work is a positive reconstruction control, not evidence that minimization succeeded.

Communication only replay is not universally useless. The result shows insufficiency for one measured single-node, four A100 decision matrix. Such replay can still help with debugging, transport isolation, or reproduction of a call sequence when the claim remains at that level.

7 Threats to validity and limitations

Single system and workload. All physical data comes from one four A100 PCIe node and one source derived workload family. GPU topology, CPU placement, software versions, thermal state, and site configuration may affect the ranking. The result does not generalize to multi-node systems, NVSwitch, other GPU generations, or vendor qualification.

Finite configuration set. The decision object contains eight NCCL configurations. It does not cover every environment variable, channel count, topology override, message size, or collective library. Adding configurations changes the number and difficulty of pair decisions.

Tie policy sensitivity. The trusted join uses an IQR rule. The exact work gate uses a pair specific absolute or relative band with bootstrap uncertainty. Both policies were frozen for their experiments, but another operational tie definition could change the labels. Exact counts and τ_b are reported so that the effect is visible.

Measurement dependence. Repeated timings within one process may be dependent. The independent percentile bootstrap is a predeclared approximation, not a universal model of system noise. A block or hierarchical bootstrap may be preferable when temporal dependence is strong. Replication across nodes and days remains future work.

Shared implementation. A defect shared by source capture, materialization, and verification could evade deterministic checks. Hash binding proves byte identity, not semantic correctness. Independent producers, differential implementations, and application invariants would provide stronger controls.

Researcher choices. Explicit selection and completeness reduce post hoc cell dropping, but campaign design, configuration choice, and policy thresholds remain researcher decisions. The exact work gate freezes thresholds before analysis. The trusted join is reported as a measured comparison, not as a formal gate defined after the result.

Publication boundary. The paper uses generated artifacts retained at a specific evidence commit. Older narrative measurements without complete evidence are excluded. This avoids unsupported reuse, although it omits historical observations that cannot now be independently reconstructed.

8 Artifact availability and reproducibility

The source repository is available at <https://github.com/iemAnshuman/CommCanary>. The paper binds evidence inspection to commit

c133ad3fb5883100cb8622b597d5b6233b5efca1.

Campaigns retain their own manifest, selection, completeness, archive, and repository identities. The outer evidence commit and the source tree used to prepare the paper are separate states.

The trusted join is under `experiments/rosteam/results/publications/`. The exact work diagnostic is under `experiments/rosteam/results/exact-work-artifacts/publication/`. The authoritative directory names are:

- `trusted-join-core-shared-overlap-primary/`;
- `decision-gate-20260801-r3-analysis-1300d12-primary/`;
- the corresponding repro directory; and
- `qualification-exact-20260730-r3-primary/`.

The supplied `scripts/` directory regenerates every vector figure from `evidence/verified_metrics.json`. That compact snapshot records verified values and their paper binding. It does not replace the immutable result directories. Reproducing physical timings still requires the recorded hardware and software environment; deterministic analysis does not imply hardware independent measurements.

No commit, push, issue, Zenodo publication, DOI reservation, or release metadata change is part of this source tree. A focused analyzer or test run must not be described as full release readiness.

9 Related work

Collective algorithms and tuning. MPI defines collective semantics but leaves the algorithm to the implementation [1]. MPICH studies show why message size and process count affect algorithm selection [3]. Performance models seek to predict or tune these choices [4]. NCCL provides GPU collectives and configurable algorithms and protocols [2]; NCCL Tests supplies operation level measurements [16]. CommCanary uses the full workload decision as the reference and tests whether a trace representation preserves it.

Trace representation and replay. ScalaTrace retains replay capability while compressing communication traces [9]. Log-GOPSim simulates trace driven communication at scale [10]. ASTRA-SIM models distributed learning workloads and networks [7, 8]. Chakra provides an execution trace graph, Mystique derives portable benchmarks from execution traces, and PARAM provides benchmark and replay machinery [5, 6, 12]. CommCanary focuses on a failure mode that these representations do not remove: a valid communication trace can still change the configuration decision when it omits causal workload context.

Compute and communication concurrency. Earlier work measures the resource interaction between computation and collective communication and develops overlap mechanisms [11]. CommCanary does not introduce another overlap engine. The experiments show that removing or changing overlap alters which NCCL configuration appears preferable in the measured case.

Reduction and workload minimization. Delta debugging removes input while preserving a predicate [17]. A future reduced canary can use a similar objective: remove trace regions while preserving a decision fidelity predicate. The publication evidence contains no completed reduced physical canary or cost result, so minimization remains a design direction.

Statistical comparison. Benchmarking guidance warns against selective statistics and incomplete variability reports [14]. We pair exact decisions with Kendall’s statistic [13] and use a predeclared bootstrap procedure [15]. Uncertainty and stability are verdict conditions rather than annotations added after a point estimate.

10 Conclusion

The 280 cell comparison gives a clear negative result for communication only replay on the measured single-node, four A100 system. W-canary reaches 57.1% pair agreement, below the isolated microbenchmark. Compute and communication concurrency changes the ranking. The strongest shared overlap replay reaches 71.4% but disagrees on eight pairs.

Exact work is much closer to the source ordering. Its 92.86% agreement, $\tau_b = 0.857$, and low relative errors pass every point threshold. The final outcome remains *inconclusive* because pair uncertainty crosses decision boundaries and one configuration violates the stability limit.

The supported contribution is an independently verifiable, fail closed evidence workflow and a measured result about the role of concurrency. Production qualification, reduced canary savings, held out regression safety, application product studies, and generalization beyond the measured system remain open.

References

- [1] Message Passing Interface Forum. MPI: A message passing interface standard, version 4.1. Technical report, Message Passing Interface Forum, November 2023. URL <https://www.mpi-forum.org/docs/mpi-4.1/mpi41-report.pdf>.
- [2] NVIDIA Corporation. NVIDIA Collective Communications Library (NCCL). Official software repository and documentation, 2026. URL <https://github.com/NVIDIA/nvcc1>. Accessed 2026-08-14.
- [3] Rajeev Thakur, Rolf Rabenseifner, and William Gropp. Optimization of collective communication operations in MPICH. *The International Journal of High Performance Computing Applications*, 19(1):49–66, 2005. doi: 10.1177/1094342005051521.
- [4] Sathish S. Vadhiyar, Graham E. Fagg, and Jack J. Dongarra. Towards an accurate model for collective communications. *The International Journal of High Performance Computing Applications*, 18(1):159–167, 2004. doi: 10.1177/1094342004041297.
- [5] Srinivas Sridharan, Taekyung Heo, Louis Feng, Zhaodong Wang, Matt Bergeron, Wenxin Fu, Shengbao Zheng, Brian Coutinho, Saeed Rashidi, Changhai Man, and Tushar Krishna. Chakra: Advancing performance benchmarking and co-design using standardized execution traces. *arXiv preprint arXiv:2305.14516*, 2023. doi: 10.48550/arXiv.2305.14516.
- [6] Mingyu Liang, Wenxin Fu, Louis Feng, Zhongyi Lin, Pavani Panakanti, Shengbao Zheng, Srinivas Sridharan, and Christina Delimitrou. Mystique: Enabling accurate and scalable generation of production AI benchmarks. In *Proceedings of the 50th Annual International Symposium on Computer Architecture, ISCA ’23*. ACM, 2023. doi: 10.1145/3579371.3589072.
- [7] Saeed Rashidi, Srinivas Sridharan, Sudarshan Srinivasan, and Tushar Krishna. ASTRA-SIM: Enabling SW/HW co-design exploration for distributed DL training platforms. In *2020 IEEE International Symposium on Performance Analysis of Systems and Software, ISPASS*, pages 81–92. IEEE, 2020. doi: 10.1109/ISPASS48437.2020.00018.
- [8] William Won, Taekyung Heo, Saeed Rashidi, Srinivas Sridharan, Sudarshan Srinivasan, and Tushar Krishna. ASTRA-sim2.0: Modeling hierarchical networks and disaggregated systems for large-model training at scale. *arXiv preprint arXiv:2303.14006*, 2023. doi: 10.48550/arXiv.2303.14006.
- [9] Michael Noeth, Prasun Ratn, Frank Mueller, Martin Schulz, and Bronis R. de Supinski. ScalaTrace: Scalable compression and replay of communication traces for high-performance computing. *Journal of Parallel and Distributed Computing*, 69(8):696–710, 2009. doi: 10.1016/j.jpdc.2008.09.001.
- [10] Torsten Hoefler, Timo Schneider, and Andrew Lumsdaine. LogGOPSIm: Simulating large-scale applications in the LogGOPS model. In *Proceedings of the 19th ACM International Symposium on High Performance Distributed Computing, HPDC ’10*, pages 597–604. ACM, 2010. doi: 10.1145/1851476.1851564.
- [11] Saeed Rashidi, Matthew Denton, Srinivas Sridharan, Sudarshan Srinivasan, Amoghavarsha Suresh, Jade Nie, and Tushar Krishna. Enabling compute communication overlap in distributed deep learning training platforms. In *Proceedings of the 48th Annual International Symposium on Computer Architecture, ISCA ’21*, pages 540–553. IEEE, 2021. doi: 10.1109/ISCA52012.2021.00049.
- [12] Meta Platforms, Inc. PARAM. Official software repository for communication benchmark and replay tooling, 2023. URL <https://github.com/facebookresearch/param>. Accessed 2026-08-14.
- [13] Maurice G. Kendall. A new measure of rank correlation. *Biometrika*, 30(1–2):81–93, 1938. doi: 10.1093/biomet/30.1-2.81.

- [14] Torsten Hoefler and Roberto Belli. Scientific benchmarking of parallel computing systems: Twelve ways to tell the masses when reporting performance results. In *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*, SC '15. ACM, 2015. doi: 10.1145/2807591.2807644.
- [15] Bradley Efron. Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1):1–26, 1979. doi: 10.1214/aos/1176344552.
- [16] NVIDIA Corporation. NCCL Tests: Performance and correctness tests for NCCL operations. Official software repository, 2026. URL <https://github.com/NVIDIA/nccl-tests>. Accessed 2026-08-14.
- [17] Andreas Zeller and Ralf Hildebrandt. Simplifying and isolating failure-inducing input. *IEEE Transactions on Software Engineering*, 28(2):183–200, 2002. doi: 10.1109/32.988498.

A Campaign identities

Table 6 lists abbreviated campaign identities. Full SHA-256 values remain in the evidence snapshot and authoritative repository artifacts. M, S, C, and R denote manifest, selection, completeness verdict, and repository commit.

Table 6: Abbreviated manifest bound campaign identities.

Run and campaign	Bound identities
core-20260724-r7	M 3bd321ec960c; S d01db822c8e9; C 46d8012b3594; R 2855275288e6
rosteam-core	M a402a4ec73ea; S a1e876861f7f; C b6cd1aae4cfb; R 2855275288e6
shared-replay-20260720-r2	M 5d5eb9f4822e; S e012e40613f5; C 5b153d0452df; R 2855275288e6
rosteam-shared-replay	M 1b1909b3e856; S c675e42e34e4; C 6377a65bb057; R 4585318ac244
overlap-20260724-r1	M c0bc2e6a9ecb; S bf23480f39dd; C 593772c09020; R 9b1e58aef91e
rosteam-overlap	
decision-gate-20260801-r3	
rosteam-decision-gate	
qualification-exact-20260730-r3	
rosteam-qualification-exact	

B Trusted join regeneration

The generated publication fragment retains the analyzer invocation with its exact argument order. The complete command is also included in this source tree at `evidence/trusted_join_regeneration.txt`. It binds the core campaign, the primary selection and completeness verdict, both joined evidence directories, the output directory, configuration identities, and median thresholds.

C Claim ledger

Supported comparative claim. The trusted join contains 280 of 280 selected successful cells. Over the measured eight configuration set, W-shared-overlap has the highest proxy agreement at 71.4%. Communication only W-canary reaches 57.1%.

Supported gate observation. Exact work has 26 of 28 point agreement, $\tau_b = 0.857$, one false negative, one false positive, 1.55% median absolute relative error, and 4.05% p95 error. The persisted outcome is *inconclusive*.

Supported diagnostic claim. One same node exact work capsule has a 1,434.112 μ s source median and a 1,541.0015 μ s replay median. All 32 deterministic checks pass. No qualification verdict is issued.

Not supported. The paper does not support production qualification, a passing decision gate, reduced canary cost savings, held out safety, real vLLM or SGLang validation, multi-node or congestion fidelity, CUDA graph fidelity, general vendor qualification, or generalization beyond the measured single-node four A100 scope.